Theory Manual

Composite Material Full Test Suite
(Tensile, Compressive, Flexural, Impact and Vibration)

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Companion document to `composite_full_test.m`

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1 Purpose and scope

This manual explains the mechanics behind each test simulated in `composite_full_test.m`: a MATLAB GUI that lets the user pick a composite material from a library, enter specimen dimensions, generate a test report, plot load–displacement and stress–strain curves, and run an animated virtual test. Seven tests are covered: tensile (ASTM D3039), drop-weight impact (D7136), Charpy (E23), Izod (D256), flexural three-point bend (D790), compressive (D695) and vibration (D3580), using the standard designations as printed in the script.

The script is a *virtual* test bench: curves come from a material library and analytical or phenomenological models, not from a load frame. Each section states the governing relations first and then how the script uses them. Where the script departs from the textbook relation this is stated, and the list is collected in Section 9.

Note on source coverage. The file received ends part-way through `animate_compressive`, and the library and helper functions (`define_material_library`, `make_stress_strain_curve`, `make_compressive_curve`, `build_report_lines`, `build_validation_lines`, `style_ax`, `make_anim_fig`, `make_dark_ax`, `compare_samples`) were not in it. Sections that depend on them describe only what the visible code does with their outputs.

2 Composite material background

A fibre-reinforced composite is anisotropic: stiffness and strength depend strongly on direction relative to the fibres. For a unidirectional ply with fibre volume fraction V_f, the simple rule-of-mixtures estimates are

$$E_1 = V_f E_f + (1 - V_f) E_m, \qquad \frac{1}{E_2} \approx \frac{V_f}{E_f} + \frac{1 - V_f}{E_m}, \qquad \rho = V_f \rho_f + (1 - V_f) \rho_m. \tag{1}$$

Longitudinal tensile strength is fibre-dominated, whereas transverse tensile, shear and compressive strengths are matrix- and interface-dominated, so a composite typically has $\sigma_{ut} \neq \sigma_{uc}$. This is why the library stores separate `uts` and `ucs` values alongside the modulus (`young_modulus`, in MPa since the GUI prints $E/10^3$ as GPa), yield and break strength, elongation, density, Poisson ratio, natural frequency and damping ratio.

3 Tensile test (ASTM D3039)

3.1 Governing relations

A straight-sided coupon with gauge length L_0 and cross-section $A = wt$ is loaded in tension at constant crosshead rate. Engineering stress and strain are

$$\sigma = \frac{F}{A}, \qquad \varepsilon = \frac{\Delta L}{L_0}, \qquad E = \left. \frac{\Delta \sigma}{\Delta \varepsilon} \right|_{\text{linear region}}, \tag{2}$$

with ultimate tensile strength $\sigma_{ut} = F_{max}/A$ and Poisson ratio $\nu = -\varepsilon_{trans}/\varepsilon_{axial}$.

3.2 Use in the script

With σ in MPa and A in mm^2, force in kN follows from $F = \sigma A/10^3$, and displacement from strain in percent, $\delta = \varepsilon_{\%} L_0/100$. The script uses exactly these two conversions to turn the stress–strain curve into the load–displacement plot. When the user enters new dimensions, the area and the ultimate, yield and break loads are recomputed as $\sigma A/10^3$ while the material strengths stay constant (a size-independent material). The animation steps through 150 frames, colours the coupon blue/amber/red at $\sigma/\sigma_{ut} < 0.5$, < 0.8 and above, applies a necking factor after 40% progress and draws crack segments after frame $0.85\,N$. Necking is a visual device: unidirectional composites typically fail without significant necking.

4 Compressive test (ASTM D695)

Compressive stress is $\sigma_c = F/A$ with strain $\varepsilon_c = \Delta h/h_0$. Compressive failure of fibre composites is governed by fibre microbuckling and kinking; the classical shear-mode estimate (Rosen) is

$$\sigma_{cu} \approx \frac{G_m}{1 - V_f}, \tag{3}$$

which overpredicts measured values because of fibre misalignment, so practical strengths are lower. Short, thick specimens suppress global (Euler) buckling, $P_{cr} = \pi^2 EI/(KL)^2$; the script's default height is $1.2L_0$ and the user can change it. The compressive curve is generated by `make_compressive_curve` (not in the received file); the animation uses the same $\delta = \varepsilon L_0/100$ and $F = \sigma A/10^3$ conversions as the tensile case.

5 Flexural test (ASTM D790, three-point bend)

For a rectangular beam of width b, depth h and support span L with central load F:

$$\sigma_f = \frac{3FL}{2bh^2}, \qquad \delta = \frac{FL^3}{48EI} \;\text{ with } I = \frac{bh^3}{12}, \qquad \varepsilon_f = \frac{6\delta h}{L^2}. \tag{4}$$

The flexural modulus from the initial slope $m = \Delta F/\Delta\delta$ is $E_f = L^3 m/(4bh^3)$. In the script, the flexural strength is *assumed* to be $\sigma_f = 1.5\,\sigma_{ut}$, the peak load is back-calculated as $F = 2\sigma_f bh^2/(3L)$ (the same relation rearranged, in N then divided by 1000 for kN), and the peak deflection from the beam formula, clipped to 0.5–15 mm for display. The load–deflection path is $F = F_{max}\,\xi^{0.9}$, $\delta = \delta_{max}\,\xi^{1.1}$ for progress $\xi \in [0, 1]$, a shaping choice rather than a material model. Span is fixed at 100 mm in the animation.

6 Impact tests

6.1 Drop-weight impact (ASTM D7136)

An impactor of mass m released from height h strikes the plate with

$$v_0 = \sqrt{2gh}, \qquad E_{in} = mgh. \tag{5}$$

The script uses $h = 1.20$ m and $m = 5.50$ kg, giving $v_0 \approx 4.85\,\mathrm{m\,s^{-1}}$ and $E_{in} \approx 64.7$ J, and a hemispherical tup of radius $R = 8$ mm (16 mm diameter, as in D7136). Contact is modelled with a Hertzian power law and the motion is integrated by explicit Euler with $\Delta t = 2\,\mu$s up to 25 ms:

$$F_c = k_h\,\delta^{3/2}, \qquad m\ddot{z} = mg - F_c, \qquad F_c \le 50\,mg, \tag{6}$$

$$v_{i+1} = v_i + a_i\Delta t, \qquad z_{i+1} = z_i + v_i\Delta t. \tag{7}$$

Absorbed energy is the area under the force–displacement curve, $E_{abs} = \int F\,\mathrm{d}\delta$, evaluated with `cumtrapz` (kN × mm = J). The true Hertz contact stiffness for a sphere on an elastic half-space is $k_h = \frac{4}{3}E^*\sqrt{R}$; the script uses $\frac{4}{3}\sqrt{R}\,E_{eff}^{1/2}$ with a floor of 2×10^7, which in practice always governs (see Section 9).

6.2 Charpy and Izod pendulum tests

In a pendulum test the absorbed energy is the loss of pendulum potential energy,

$$E_{abs} = mgR\,(\cos\beta - \cos\alpha), \tag{8}$$

where α is the release angle and β the follow-through angle. Charpy supports the specimen as a simply supported beam (struck at mid-span); Izod clamps it as a vertical cantilever and strikes it above the clamp. The script does not integrate the pendulum; it estimates a strain-energy-like value and clips it:

$$E_{Charpy} = \mathrm{clip}\left(0.5\,t\,w\,\frac{(\sigma_{ut}/10^3)^2}{E}\,10^6,\ 5,\ 150\right)\ \mathrm{J}, \tag{9}$$

$$E_{Izod} = \mathrm{clip}\left(0.40\,t\,w\,\frac{(\sigma_{ut}/10^3)^2}{E}\,10^6,\ 2,\ 80\right)\ \mathrm{J}. \tag{10}$$

These follow the shape of the strain energy $U \propto \sigma^2 V/(2E)$ but are heuristic, and the clip limits dominate for many inputs. The swing animation (release $-145°$ to $+75°$ for Charpy, $-80°$ to $+55°$ for Izod) is illustrative and not tied to E_{abs} through mgR.

7 Vibration test (ASTM D3580 as labelled)

7.1 Single-degree-of-freedom model

The specimen is represented by a damped oscillator with natural frequency f_n and damping ratio ζ. With frequency ratio $r = f/f_n$ the dynamic amplification (magnification factor) is

$$H(r) = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad H_{max} \approx \frac{1}{2\zeta} \text{ at } r \approx 1. \tag{11}$$

The half-power (bandwidth) method recovers damping from a measured response: $\zeta \approx (f_2 - f_1)/(2f_n)$, where f_1, f_2 are the frequencies at $H_{max}/\sqrt{2}$. The script plots $H(r)$ for the selected material and runs a swept-sine test from 5 to 200 Hz over 200 frames, scaling amplitude as $1.5\,H$ mm and flagging resonance when $|r - 1| < 0.1$.

7.2 Link to beam properties

For a uniform beam, the first bending frequency is

$$f_1 = \frac{\lambda_1^2}{2\pi}\sqrt{\frac{EI}{\rho A L^4}}, \qquad \lambda_1 = 1.875 \text{ (cantilever)}, \quad \pi \text{ (simply supported)}, \tag{12}$$

showing that stiffness-to-density ratio sets frequency; the script takes f_n from the library rather than computing it from the entered dimensions, so changing dimensions does not shift the resonance.

8 Software architecture

- **Main function and nested callbacks.** All callbacks are nested in `composite_full_test`, sharing the workspace; application state is stored with `guidata(fig)`.
- **Start-up.** `inputdlg` collects tester name, company and reference number.
- **Static view.** `on_sample_change` rebuilds the report, the load–displacement, stress–strain and frequency plots, the validation list and the status bar.
- **Animation path.** `run_animation` → `get_user_dimensions` → copy of the material struct with new width, thickness, length and recomputed area and loads → one `animate_*` function per test.
- **Unit conventions.** MPa for stress, mm for lengths, mm^2 for area, kN for load, MPa for modulus, J for energy.

9 Assumptions, limitations and suggested corrections

1. **Cancelling the dimensions dialog fails.** `get_user_dimensions` returns a numeric array on cancel, but `run_animation` reads `dims{1}`; use a cell array, `dims = {default_width, default_thickness, default_gauge};`.
2. **User flexural span is ignored.** The span entered is stored in `gauge_length` but `animate_flexural` fixes `L_span = 100`; use `L_span = d.gauge_length`.
3. **Drop-weight contact stiffness is material-independent.** $\frac{4}{3}\sqrt{R}\,E^{1/2}$ is of order 10^4–10^5 for realistic moduli, so the 2×10^7 floor always applies. Use $k_h = \frac{4}{3}E^*\sqrt{R}$ with E^* in Pa (or a measured contact law) if material sensitivity is wanted.
4. **Charpy and Izod energies are heuristic** and clipped; they are not $mgR(\cos\beta - \cos\alpha)$ from a pendulum balance.

5. **Flexural strength** is assumed $1.5\,\sigma_{ut}$ instead of a library value.

6. **Frequency response and time trace.** $H(r)$ is the force-excitation magnification factor; base excitation has transmissibility $\sqrt{1+(2\zeta r)^2}\,H(r)$. The decaying time signal uses the excitation frequency in the exponent instead of ω_n.

7. **Standards labelling.** ASTM E23 covers notched-bar impact of metallic materials (plastics and composites are usually tested under D6110 or D256 as appropriate), and D3580 is a packaging vertical-vibration method; modal or damping tests of composites more often follow other methods (for example E756). Verify the designations before citing them in a report.

8. **Hard-coded report date** ("4/16/2026") in the header.

9. **Visual-only effects.** Necking, crack lines and colour changes are illustrative.